

Quick Check

Lesson 3: Carbohydrates: Starch and Sugars

7 multiple-choice questions • choose A, B, C or D

Sub-Strand 1.2: Chemicals of Life | Lesson 3

A group adds 2-3 drops of iodine solution to some maize flour mixed with water. Which result shows that starch is present?

- A** The mixture turns brick-red
- B** The mixture turns blue-black
- C** The mixture turns yellow-orange and stays clear
- D** The mixture turns green

A learner mixes 2 cm³ of mashed ripe banana solution with 2 cm³ of Benedict's solution and heats it in a hot water bath for 4 minutes. It turns orange to brick-red. What does this show?

- A** The banana contains starch
- B** The banana contains a lot of reducing sugar
- C** The banana contains protein
- D** The banana contains no carbohydrate

Ugali turns blue-black with iodine solution but stays blue when heated with Benedict's solution. What does this tell us about the carbohydrate in ugali before digestion?

A It contains starch but no reducing sugar such as glucose

B It contains no carbohydrate

C It contains both starch and a lot of glucose

D It contains glucose but no starch

Ugali has no reducing sugar, yet Wanjiru's blood glucose went up after she ate it. Which statement best explains this?

- A** Starch is already glucose, so it passes straight into the blood
- B** Beans turn into starch in the stomach, which raises the glucose
- C** Starch changes into protein, which then enters the blood as glucose
- D** Starch is made of many glucose units joined together, so it must be broken down into glucose before entering the blood

Wanjiru's blood glucose concentration was 4.5 mmol/L before lunch and reached a peak of 8.0 mmol/L after 60 minutes. By how much did it rise, in mmol/L?

A 3.0

B 3.5

C 12.5

D 4.5

After today's food tests, the class DQB has these four questions. Which one can the class now answer using the test results?

- A** Does the body use only carbohydrates for energy?
- B** Why did Wanjiru's blood glucose fall after 60 minutes?
- C** Which enzyme breaks starch into glucose?
- D** Do ugali and maize flour contain starch?

Which statement about the chain from ugali to blood glucose is supported by today's evidence and should be added to the 'food chemicals' box of the class model?

- A** Ugali contains starch, which is made of glucose units, so it can be a source of the glucose in the blood
- B** Nyama choma gives a blue-black result with iodine, so it is the main source of starch in Wanjiru's lunch
- C** Ugali gives a brick-red result with Benedict's solution, so it already contains glucose
- D** Only sweet foods contain carbohydrates, so ugali cannot be a source of glucose

All done

Ask your teacher how to submit your answers today (for example, on a paper answer slip).